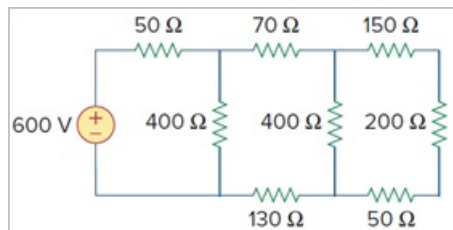


Problem

Using series/parallel resistance combination, find the equivalent resistance seen by the source in the circuit of Fig. 2.98. Find the overall absorbed power by the resistor network.

Figure 2.98:



Step-by-step solution

Step 1 of 4

Consider the circuit diagram of textbook Figure 2.98. The resistors $150\ \Omega$, $200\ \Omega$ and $50\ \Omega$ to the right of the circuit are in series. Therefore their equivalent resistance is,

$$150 + 200 + 50 = 400\ \Omega \quad \dots\dots (1)$$

The resistor $400\ \Omega$ in the right of the circuit and $400\ \Omega$ resistor of equation (1) are in parallel. Therefore their equivalent resistance is,

$$\begin{aligned} R_{eq1} &= (400) \parallel (400) \\ &= \frac{(400)(400)}{800} \\ &= 200\ \Omega \end{aligned}$$

Step 2 of 4

The resistors $70\ \Omega$, R_{eq1} and $130\ \Omega$ are in series and their resultant is in parallel to $400\ \Omega$. Therefore their equivalent resistance is,

$$\begin{aligned} R_{eq2} &= (70 + R_{eq1} + 130) \parallel 400 \\ &= (70 + 200 + 130) \parallel 400 \\ &= (400) \parallel (400) \\ &= 200\ \Omega \end{aligned}$$

The equivalent resistance of the circuit is,

$$\begin{aligned} R_{eq3} &= 50 + R_{eq2} \\ &= 50 + 200 \\ &= 250\ \Omega \end{aligned}$$

Therefore, the equivalent resistance seen by the source is $\boxed{250\ \Omega}$.

Step 3 of 4

The modified circuit is as shown in Figure 1.

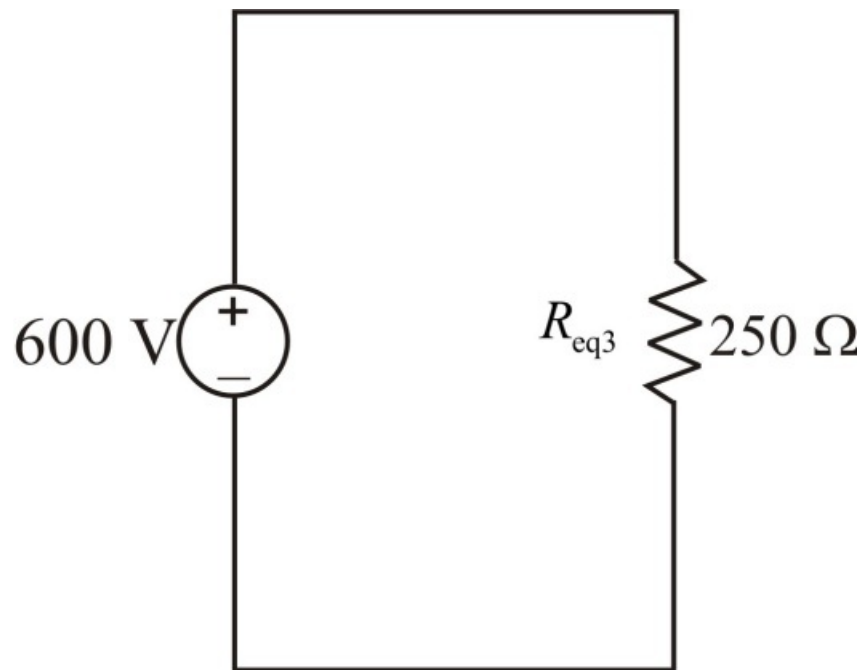


Figure 1

Step 4 of 4

The overall power absorbed by the circuit is,

$$\begin{aligned} P &= \frac{V^2}{R_{eq3}} \\ &= \frac{(600)(600)}{250} \\ &= 1.44 \text{ kW} \end{aligned}$$

Therefore, the overall power absorbed by the circuit is 1.44 kW.